

ShipSafe AI

Automated Release Safety Engineer

IBM Bob Hackathon Submission

Built for the Internet of Agents

The Problem: The DevOps Crisis

- CI/CD pipelines automate building & testing, but **security and architectural safety** still rely on human oversight.
- Senior engineers are overwhelmed by manual code reviews.
- Cognitive load is high, and tight deadlines mean code is merged with undetected structural flaws.
- **The Cost:** Catastrophic production outages, severe data breaches, and loss of user trust.

The Solution: ShipSafe AI

ShipSafe AI is an **autonomous, highly intelligent Release Safety Engineer** that sits directly in the developer's workflow.

1. **Active Participant:** Not just a passive scanner.
2. **Deep Analysis:** Detects critical risks with deterministic accuracy in real-time.
3. **Visual Simulation:** Simulates how failures manifest in production.
4. **Automated Remediation:** Hands off vulnerabilities to **IBM Bob** to automatically write code refactors.

Target Users & The Workflow

Users: DevOps Engineers, Release Managers, DevSecOps, Full-Stack Developers.

The Workflow:

1. **Upload:** Drop codebase into the Analysis Workspace.
2. **Scan:** Custom backend calculates a dynamic "Project Risk Score".
3. **Triage:** Interactive Results Dashboard highlights vulnerable code.
4. **Simulate:** Watch a visual timeline of a cascading production outage.
5. **Remediate:** Launch the AI Remediation Workspace to fix it instantly.

The Core Intelligence: IBM Bob

ShipSafe AI integrates **IBM Bob** as our autonomous, highly-skilled Senior Security Pair-Programmer.

- **The Handoff:** ShipSafe securely packages the vulnerable code, file context, and exact security risk profile.
- **The Prompt:** IBM Bob adopts the persona of a strict enterprise security architect (adhering to OWASP guidelines).
- **The Fix:** IBM Bob analyzes the flawed logic and generates a clean, secure, and fully refactored code-diff.

IBM Bob: Specific Use-Cases

- **SQL Injection Remediation:** Converts dangerous string interpolation into secure parameterized statements.
- **Secrets Management:** Refactors hardcoded API keys to securely import from environment variables.
- **Resilience & Reliability:** Rewrites infinite `while(true)` loops into bounded `for` loops with exponential backoff and timeout parameters.

Creativity and Uniqueness

- **Visually Stunning:** Premium Next.js, Tailwind CSS, and Framer Motion UI.
- **Interactive Core:** Custom "AI Release Safety Core" glassmorphism node-graph.
- **Closing the Loop:** Traditional scanners only point out problems. ShipSafe AI *shows* you the failure visually and uses IBM Bob to *write* the fix.
- **Engineering Experience:** Turns a dry compliance task into a highly engaging, productive workflow.

Thank You!

ShipSafe AI

Ready for Deployment.